$$\underline{x}(t) = \Phi(t,t_0)\underline{x}_0 + \int_{t_0}^{t} \Phi(t,\tau)B(\tau)\underline{u}(\tau)\,d\tau$$
$$\underline{x}(t_0) = \underline{x}_0$$

For LTI:

$$\Phi(t,\tau) = e^{(t-\tau)A} = \sum_{k=0}^{\infty} \frac{(t-\tau)^k}{k!} A^k$$

Note: since STM for LTI depends only on elapsed time, $t-\tau$, can w.l.o.g. Assume $t_0 = 0$

So $$\underline{x}(t) = e^{At}\underline{x}_0 + \int_0^t e^{A(t-\tau)} B\,\underline{u}(\tau)\,d\tau$$

[Note $e^{A(t-\tau)}$ calculated by replacing "t" in expansion of e^{At} with "$t-\tau$" i.e scalars $e^{at} \to e^{a(t-\tau)}$]

So how to explicitly expand e^{At}?

Common e^{At}:

① $A = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} = \text{diag}\{a_1, a_2\}$

$$e^{At} = \begin{bmatrix} e^{a_1 t} & 0 \\ 0 & e^{a_2 t} \end{bmatrix} = \text{diag}\{e^{a_1 t}, e^{a_2 t}\}$$

$A = \text{diag}\{a_1, \ldots, a_n\}$

$\Rightarrow e^{At} = \text{diag}\{e^{a_k t}\}_{k=1}^{n}$

② $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

$$e^{At} = e^{at}\begin{bmatrix} \cos bt & \sin bt \\ -\sin bt & \cos bt \end{bmatrix}$$

③ $A = \begin{bmatrix} a & 1 \\ 0 & a \end{bmatrix}$

$$e^{At} = e^{at}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

3.) extends to n×n

$$A = \begin{bmatrix} a & 1 & 0 & \cdots & 0 \\ 0 & a & 1 & 0 \cdots & 0 \\ & & \ddots & & \vdots \\ & \emptyset & & \ddots & 0 \\ & & & & 1 \\ & & & & a \end{bmatrix}$$

Then

$$e^{At} = e^{at} \begin{bmatrix} 1 & t & \frac{1}{2}t^2 & \cdots & \frac{1}{(n-1)!}t^{n-1} \\ 0 & 1 & t & \frac{1}{2}t^2 \cdots & \frac{1}{(n-2)!}t^{n-2} \\ & & \ddots & & \vdots \\ 0 & \cdots & & & t \\ 0 & \cdots & & & 1 \end{bmatrix}$$

= Add'l property:

if $A = \text{blkdiag}\{A_1, \cdots, A_r\}$

then

$$e^{At} = \text{blkdiag}\{e^{A_1 t} \cdots e^{A_r t}\}$$

each A_k is $n_k \times n_k$

with $\sum_{k=1}^{r} n_k = n$

c.c.

$$A = \begin{bmatrix} A_1 & & & \\ & A_2 & \emptyset & \\ & \emptyset & \ddots & \\ & & & A_r \end{bmatrix}$$

(A_1 is $n_1 \times n_1$, A_2 is $n_2 \times n_2$, A_r is $n_r \times n$, overall $n \times n$)

$$e^{At} = \begin{bmatrix} e^{A_1 t} & & & \\ & e^{A_2 t} & & \\ & & \ddots & \\ & & & e^{A_r t} \end{bmatrix}$$

Example: A 5×5

$$A = \begin{bmatrix} \overbrace{\begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}}^{A_1} & 0 & 0 & 0 \\ 0 & 0 & \overbrace{\begin{bmatrix} -1 & 3 \\ -3 & -1 \end{bmatrix}}^{A_2} & 0 \\ 0 & 0 & 0 & \underbrace{[-3]}_{A_3} \end{bmatrix}$$

$$= \text{blkdiag}\{A_1, A_2, A_3\}$$

$$e^{At} = \text{blkdiag}\{e^{A_1 t}, e^{A_2 t}, e^{A_3 t}\}$$

$$= \begin{bmatrix} \begin{bmatrix} e^{-2t} & te^{-2t} \\ 0 & e^{-2t} \end{bmatrix} & 0 & 0 & 0 \\ 0 & 0 & \begin{bmatrix} e^{-t}\cos 3t & e^{-t}\sin 3t \\ -e^{-t}\sin 3t & e^{-t}\cos 3t \end{bmatrix} & 0 \\ 0 & 0 & 0 & \begin{bmatrix} e^{-3t} \end{bmatrix} \end{bmatrix}$$

More systematically:

One approach: use Laplace!

$$F(s) = \mathcal{L}\{f\} \triangleq \int_{0^-}^{\infty} e^{-st} f(t)\, dt$$

Recall important properties

Linearity: if a_1, a_2 constant,

$$\mathcal{L}\{a_1 f_1 + a_2 f_2\} = a_1 F_1 + a_2 F_2$$

Warning: this does NOT hold if a_1, a_2 are themselves time-varying

Derivative

$$\mathcal{L}\{\dot{f}\} = sF(s) - f(0^-)$$

Apply to vectors as well

$$\mathcal{L}\{x(t)\} = \begin{bmatrix} \mathcal{L}\{x_1(t)\} \\ \mathcal{L}\{x_2(t)\} \\ \vdots \\ \mathcal{L}\{x_n(t)\} \end{bmatrix} = \begin{bmatrix} X_1(s) \\ X_2(s) \\ \vdots \\ X_n(s) \end{bmatrix}$$

$$= X(s)$$

$$\mathcal{L}\{\dot{\underline{x}}(t)\} = \begin{Bmatrix} sX_1(s) - x_1(0^-) \\ sX_2(s) - x_2(0^-) \\ \vdots \\ sX_n(s) - x_n(0^-) \end{Bmatrix}$$

$$= s\underline{X}(s) - \underline{x}_0$$

$$= s\mathbb{I}\underline{x}(s) - \underline{x}_0$$

with $\underline{x}_0 = \begin{bmatrix} x_1(0^-) \\ \vdots \\ x_n(0^-) \end{bmatrix}$

Linear holds in linear alg. sense

$$\mathcal{L}\{A\underline{x}(t)\} = A\underline{x}(s)$$

if A is constant

Apply $\mathcal{L}$T to homog. eq'n

$$\mathcal{L}\{\dot{\underline{x}}(t) = A\,\underline{x}(t)\}$$

$$s\mathbb{I}\underline{x}(s) - \underline{x}_0 = A\,\underline{x}(s)$$

$$\Rightarrow \quad \underline{x}(s) = \underbrace{[s\mathbb{I} - A]^{-1}}_{\Phi(s)}\underline{x}_0$$

$\Phi(s)$ "resolvent matrix"

$$\underline{x}(s) = \underline{\Phi}(s)\underline{x}_0$$

$$\underline{x}(t) = \mathcal{L}^{-1}\{\underline{\Phi}(s)\underline{x}_0\}$$

$$= \mathcal{L}^{-1}\{\Phi(s)\}\underline{x}_0$$

=

But we know

$$\underline{x}(t) = e^{At}\underline{x}_0$$

$$\Rightarrow e^{At} = \mathcal{L}^{-1}\{\Phi(s)\}$$

$$= \mathcal{L}^{-1}\{(s\mathbb{I} - A)^{-1}\}$$

Note:

$$(sI - A)^{-1} = \frac{Q(s)}{r(s)}$$

$$Q(s) = \text{Adj}(sI - A)$$

$n \times n$ matrix of polynomials in s
max degree of each poly is $n-1$

$$r(s) = \text{Det}(sI - A)$$

scalar poly in s of degree n.

$\Rightarrow$ Each entry of $(sI-A)^{-1}$ is a ratio of polynomials in s.

$\Rightarrow$ Each ratio has same denominator.

$\Rightarrow$ find e^{At} by inverse xforming each of these rational fns.

$\Rightarrow$ use partial fractions

Example:

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}$$

$$(sI - A) = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}$$

$$Q(s) = \begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}$$

$$r(s) = s(s+5) + 6 = s^2 + 5s + 6 = (s+2)(s+3)$$

$$e^{At} = \mathcal{L}^{-1}\begin{bmatrix} \frac{s+5}{(s+2)(s+3)} & \frac{1}{(s+2)(s+3)} \\ \frac{-6}{(s+2)(s+3)} & \frac{s}{(s+2)(s+3)} \end{bmatrix}$$

Look at (1,1) entry

$$\frac{s+5}{(s+2)(s+3)} = \frac{c_1}{s+2} + \frac{c_2}{s+3}$$

$$\Rightarrow \mathcal{L}^{-1}\{\cdot\} = c_1 e^{-2t} + c_2 e^{-3t}$$

$$c_1 = \left[(s+2)\left(\frac{s+5}{(s+2)(s+3)}\right) \right]_{s=-2}$$

$$= \frac{3}{1} = 3 = c_1$$

$$c_2 = \left[(s+3)\left(\frac{s+5}{(s+2)(s+3)}\right) \right]_{s=-3}$$

$$= \frac{2}{-1} = -2 = c_2$$

(1,1) entry of e^{At} here is

$$\Rightarrow 3e^{-2t} - 2e^{-3t}$$

Repeat for other entries (exercise!)

$$A = \begin{bmatrix} 0 & 1 \\ -5 & -6 \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ 6(e^{-3t} - e^{-2t}) & 3e^{-3t} - 2e^{-2t} \end{bmatrix}$$

$\Rightarrow$ every entry of e^{At} is a linear comb. of the same 2 scalar exponential fns e^{-2t}, e^{-3t}

$\Rightarrow$ B/c each inverse transform has same Denominator: $(s+2)(s+3)$

equiv $\{e^{\lambda_k t}\}$, λ_k eigenvalue of A

Conclude: Generally each entry of e^{At} will be a linear comb of the scalar exponential functions $\{e^{p_k t}\}$ where p_k are the roots of char. poly

$$\Gamma(s) = \text{Det}(sI - A)$$

$\Rightarrow$ these roots are the Poles of the system.

But note

$$\Gamma(s) = 0 = \text{Det}(sI - A)$$

also define the eigenvalues of A

= Note, will also get terms like $t^l e^{\lambda_k t}$ if poles/eigenvalues repeated, like

$$\Gamma(s) = (s+1)(s+2)^2$$

$\Rightarrow$ will treat repeated pole case separately later.

=

For complete system, Laplace gives:

$$\mathcal{L}\{\dot{\underline{x}} = A\underline{x} + B\underline{u}\}$$

$$\mathcal{L}\{\underline{y} = C\underline{x} + D\underline{u}\}$$

$$(sI - A)\underline{x}(s) = \underline{x}_0 + B\underline{u}(s)$$

$$\underline{u}(s) = \mathcal{L}\{\underline{u}(t)\}$$

$$\underline{Y}(s) = C\underline{x}(s) + D\underline{u}(s)$$

=

$$\underline{x}(s) = \underbrace{(sI - A)^{-1}\underline{x}_0}_{\underline{x}_{zi}(s)} + \underbrace{(sI - A)^{-1}B\underline{u}(s)}_{\underline{x}_{zs}(s)}$$

$$\underline{x}(t) = \mathcal{L}^{-1}\{\underline{X}(s)\}$$

$$= \underline{x}_{zi}(t) + \underline{x}_{zs}(t)$$

$$\underline{x}_{zi}(t) = \mathcal{L}^{-1}\{(sI - A)^{-1}\underline{x}_0\}$$

$$= e^{At}\underline{x}_0$$

$$\underline{x}_{zs}(t) = \mathcal{L}^{-1}\{\overbrace{(sI - A)^{-1}}^{Q(s)/r(s)} B\,\underline{u}(s)\}$$

$$= \mathcal{L}^{-1}\left\{\left(\frac{Q(s)B}{r(s)}\right)\underline{u}(s)\right\}$$

$$=$$

$$\underline{y}(s) = C\underline{x}(s) + D\underline{u}(s)$$

$$= \overbrace{C(sI - A)^{-1}\underline{x}_0}^{y_{zi}(t)} + \overbrace{\underbrace{[C(sI - A)^{-1}B + D]}_{G(s)}\underline{u}(s)}^{y_{zs}(t)}$$

$$G(s) = C(sI - A)^{-1}B + D$$

p x m transfer f'n matrix for system (TFM)

We Know

$$y_{zs}(t) = \mathcal{L}^{-1}\{G(s)\underline{u}(s)\}$$

$$= \int_0^t \underbrace{G(t-\tau)}_{\text{impulse response matrix}}\underline{u}(\tau)\,d\tau$$

$$\Rightarrow G(t) = \mathcal{L}^{-1}\{G(s)\}$$

$$= Ce^{At}B + D\delta(t)$$

p x m impulse response matrix is inverse xform of TFM.

=

Remember: Above is for LTI only! Laplace breaks down in LTV since

$$\mathcal{L}\{A(t)\underline{x}(t)\} \neq A(t)\underline{X}(s)$$

similarly $\neq \mathcal{L}\{A(t)\}\underline{X}(s)$

and same holds for $\mathcal{L}\{B(t)\underline{u}(t)\}$

etc.